

Hai Zhou

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Research Interests: Recursive Self-Improvement; Verifiable Reasoning; Reliable AI Agents; Multimodal Grounding; Learning from Verified Experience

EDUCATION

University College London

Sep 2025 – Sep 2026

M.Sc. Integrated Machine Learning Systems

London, UK

Current weighted average: **78.06/100** (Distinction-level); final classification pending.

Selected marks: *Applied Machine Learning Systems I/II* (84/85); *Deep Learning for NLP* (85).

Nanjing University

Sep 2021 – Jul 2025

B.Eng. Microelectronic Science and Engineering

Nanjing, China

GPA: 4.36/5.0; average score: **87.2/100**. Undergraduate thesis: VO₂-based terahertz reconfigurable leaky-wave antenna (93/100).

RESEARCH EXPERIENCE

Auditable Compositional Question Answering over UK Public Procurement [\[GitHub\]](#)

May 2026 – Present

M.Sc. Thesis · Manuscript in preparation · Supervised by Prof. Miguel Rodrigues

Knowledge Graphs · Typed Program Synthesis · Deterministic Verification · Self-Training

- Designed an **auditable compositional QA system** for real-world public procurement, replacing top-*k* RAG with typed program synthesis and exhaustive deterministic execution for multi-hop joins, set operations and aggregation; every answer is returned with supporting records or an explicit typed abstention rather than free-form generation.
- Built the underlying data and evaluation stack from **166,277 UK procurement releases**, resolving fragmented organisation identities and monetary semantics into a provenance-preserving knowledge graph and generating **12,828 program-first QA tasks** with executable gold programs, evidence and abstention labels; an independently implemented evaluator reproduced the oracles at **99.88% agreement**.
- Developed a **generate–verify–train pipeline** in which teacher or student models propose programs in a closed **17-operator typed algebra**, while grammar constraints, grounding, deterministic execution and oracle agreement determine which traces become training signals; enabled **self-harvesting** of verified supervision from the student without continued teacher access.
- Achieved **85.65%** on a sealed 2,285-question test with a fully local 8B model versus **69.76% for the cloud teacher**, and **78.31% vs. 50.33%** on a generator-independent compositional challenge set; transferred the executable framework to **WikiTableQuestions**, improving an untrained **22.51%** baseline to **51.80%**.

Multimodal Document Reranking with Vision-Language Models [\[GitHub\]](#)

Jan 2026 – Mar 2026

Research Project · University College London

Qwen2.5-VL-7B · ColQwen2 · SFT · GRPO · LoRA · Distributed Training

- Designed an end-to-end **multimodal reranking pipeline** for RAG, using **ColQwen2**, a **late-interaction vision retriever**, to retrieve candidate document pages and a Qwen2.5-VL reasoning model to perform second-stage ranking over text, tables, figures, and complex layouts.
- Built a quality-controlled training pipeline from MMDocIR: mined **hard negatives** from retriever scores, generated relevance reasoning for each candidate page independently to avoid multi-image reasoning shortcuts, and refined redundant traces into concise CoT supervision, producing **7.2K high-quality SFT samples**.
- Developed a two-stage **SFT → GRPO** framework with a Top-1-focused ranking reward and structured-format reward; constructed **3K resolution-balanced RL samples** to reduce optimization variance from heterogeneous visual-token and image-resolution distributions.
- Trained Qwen2.5-VL-7B with **distributed mixed-precision training on 4× NVIDIA A100 GPUs**; achieved **69.5% Macro Recall@1 on MMDocIR**, improving ColQwen2 from **64.8% to 69.5% (+4.7 pp)** and outperforming a Qwen2.5-VL-32B CoT reranker.

GUI Element Localisation with Vision-Language Models [\[GitHub\]](#)

Sep 2025 – Nov 2025

Course Research Project · University College London

Qwen2.5-VL-3B · Multimodal Grounding · DeepSpeed · Distributed VLM Training

- Designed a **DOM-independent cross-platform GUI grounding system** mapping screenshots and natural-language instructions directly to clickable pixel coordinates across desktop, web and mobile interfaces, avoiding brittle XPath/CSS selectors and recorded coordinates.
- Engineered a unified pipeline over **~24K multi-platform UI screenshots**, converting point/bounding-box annotations into multi-turn grounding dialogues; reused each screenshot for up to **30 localisation turns**, yielding up to **~720K element-level grounding interactions** with answer-only supervision.
- Developed **resolution-aware coordinate alignment** around Qwen2.5-VL's dynamic resizing, mapping labels into model-space for training and generated coordinates back to original-screen space at inference, preventing localisation drift across **720p–4K** resolutions.
- Trained Qwen2.5-VL-3B with **DeepSpeed ZeRO-2 and BF16**; achieved **84.9% on ScreenSpot** and **86.4% on ScreenSpot-v2**, improving the base model by **29.4 pp** on ScreenSpot and surpassing **UI-TARS-2B** by **2.6 / 1.7 pp**.

SELECTED RESEARCH PROJECTS

Physics-Constrained Vision Models for Pasture Biomass Prediction under Geographic Domain Shift [GitHub] Apr 2026 – Jun 2026

Applied Machine Learning Systems II · University College London · 85/100

- Built a **physics-constrained vision regression framework** that predicts independent biomass components and algebraically derives coupled targets, guaranteeing $GDM = Green + Clover$ and $Total = GDM + Dead$ at inference; compared ResNet-50, EfficientNet-B0 and ViT-Small with multi-task, metadata-fusion and zero-inflated variants.
- ViT-Small achieved 0.556 ± 0.023 **weighted R^2 IID**; under a held-out Tasmania split it fell to -0.005 , while a metadata-only Ridge baseline reached $+0.251$, revealing severe geographic visual domain shift.

KG-GNN-DPO: Knowledge Graph-Grounded Preference Optimisation for Multi-Hop QA [GitHub] Apr 2026 – Jun 2026

Deep Learning for NLP · University College London · 85/100

- Developed a **training-time KG grounding pipeline**: a 2-layer R-GCN scores structural reachability and a DeBERTa NLI cross-encoder scores semantic consistency; fused the signals to construct **387 human-annotation-free preference pairs** for Qwen2-0.5B, with **no KG retrieval at inference**.
- Achieved the best results among five controlled systems on MetaQA 3-hop (**5.27% EM, 5.91% F1, 44.90% KG path consistency**), improving EM over SFT by **25.5% relative**; random-label DPO underperformed SFT, showing preference optimisation amplifies signal quality as well as noise.
- Observed a **44.90% structural-consistency vs. 3.93% NLI-faithfulness gap**, showing graph reachability and textual entailment capture distinct dimensions of grounding in entity-centric KGQA.

EARLIER RESEARCH & ENGINEERING

VO₂-Based Terahertz Reconfigurable Leaky-Wave Antenna Sep 2024 – Jun 2025

Undergraduate Thesis · Nanjing University · 93/100 · Supervised by Prof. Jingbo Wu

- Designed a **1-THz reconfigurable leaky-wave antenna** integrating substrate-integrated image-guide structures, metasurface phase modulation and VO₂ phase-change control; evaluated electromagnetic and electro-thermal behaviour using **CST and COMSOL**.
- Demonstrated **programmable beam steering in simulation**, with radiation directions consistent with generalized Snell's-law predictions.

Design and Fabrication of a Single-Qubit Superconducting Quantum Chip 2023 – 2025

Provincial Innovation & Entrepreneurship Training Grant · Nanjing University

- Completed device design, fabrication-process planning and experimental preparation for a **single-qubit superconducting chip**, combining circuit/device modelling with micro/nano-fabrication workflow design in an interdisciplinary student research team.

TECHNICAL SKILLS

Programming & Systems: Python, SQL, JavaScript/TypeScript, Linux, Git, Docker, Pandas, Parquet, CUDA.

ML & LLM Systems: PyTorch, Transformers, DeepSpeed, TRL, PEFT/LoRA, vLLM, SFT, GRPO, DPO, constrained decoding, vision-language models, knowledge graphs.

Reasoning & Evaluation: Semantic parsing, deterministic execution, benchmark design, leakage-controlled splits, McNemar tests, bootstrap confidence intervals.

Scientific Computing: MATLAB, CST Studio Suite, COMSOL Multiphysics.

SELECTED HONOURS & AWARDS

National Second Prize, Chinese Collegiate Computing Competition 2023

Honorable Mention, Mathematical Contest in Modeling (MCM) 2024

Provincial Innovation & Entrepreneurship Training Grant 2023 – 2025

Renmin Scholarship, Nanjing University – three consecutive years 2022 – 2024

LEADERSHIP & SERVICE

Captain, Electronics Debate Team; Member, Nanjing University Debate Team – led a 13-member team in case development and training; won the 2024 School Cup Debate Championship. Sep 2022 – Sep 2024

Head, Organisation Department, School Student Union – coordinated nearly ten school-level activities across planning, cross-team communication, promotion and on-site delivery. Sep 2022 – Sep 2024

Online Volunteer Tutor, Middle-School Physics – provided sustained one-to-one tutoring, diagnosing reasoning gaps and guiding step-by-step problem solving. Feb 2025 – Jun 2025